

Coding Task: EK Model

Start by reading [EK \(2002\)](#).

This assignment guides your through the construction of a simple quantitative model based on EK (2002), its calibration, and its implementation.

1 Counterfactuals in EK

We start with the simplest version of the EK model seen in class augmented to have trade imbalances. Specifically, consider the same environment of EK (2002) introduced in week 2 that features multiple countries, a single sector, and production functions that use only labor. We then extend the model to allow for trade imbalances: total spending in country n now includes international transfers,

$$X_n = w_n L_n + h(w) D_n, \quad (1)$$

where D_n is an international transfer such that $\sum_{n=1}^N D_n = 0$. The function $h(w)$ is the numeraire of such transfers. For example, if $h(w) = w_{\text{US}}$, then transfers are denominated in terms of US wages. Importantly, this numeraire function is the same for every country.

- i. Derive the exact hat algebra version of the non-linear system of equations that determines changes in wages as a function of shocks in trade costs ($\hat{d}_{ni}$), productivity ($\hat{T}_n$), and transfers ($\hat{D}_n$). Your system should be written in terms of the trade elasticity θ , and the initial matrices of trade flows X_{ni}^0 . Use this system to show that the excess labor demand for the change in wages is given by

$$Z_i(\hat{w}) \equiv \sum_{n=1}^N \frac{X_{ni}^0}{\sum_{d=1}^N X_{di}^0} \frac{\hat{T}_i (\hat{d}_{ni} \hat{w}_i)^{-\theta}}{\sum_o \pi_{no}^0 \hat{T}_o (\hat{d}_{no} \hat{w}_o)^{-\theta}} \left(y_n^0 \hat{w}_n \hat{L}_n + (1 - y_n^0) \frac{h(w^0 \hat{w})}{h(w^0)} \hat{D}_n \right) - \hat{w}_i \hat{L}_i \quad (2)$$

where $y_n^0 = \frac{\sum_{i=1}^N X_{in}^0}{\sum_{i=1}^N X_{ni}^0}$ is the income-to-spending ratio of country n .

- ii. Download the file “bilateral_trade_country.csv” from this [link](#). It was built from the 2014 WIOT Table from [WIOD 2016 Release](#). The variable trade_flow2014 is the value of trade flows, $\tilde{X}_{ni}$, from an origin country i (country_org) to another destination country n (country_dest). Use this data to calibrate the matrix of bilateral trade

flows between countries in the model, X_{ni}^0 , and compute the vector of income-to-spending ratios, y_n^0 . Calibrate the trade elasticity to four: $\theta = 4$.

- iii. Assume that transfers are set in terms of the US wage (i.e., $h(w) = w_{\text{US}}$), and set the US wage to be the numeraire (i.e., $w_{\text{US}}^0 = \hat{w}_{\text{US}} = 1$). Compute the impact of a 10% increase in Chinese productivity, $\hat{T}_{\text{China}} = 1.1$, on the wage (relative to the US wage) of each country. You can do so with the following iterative algorithm:

- (a) Consider steps, $b = 1, \dots, B$. Guess that the solution is $\hat{w}_i(b = 0) = 1$ for all $i \neq \text{US}$.
- (b) Given a guess $\hat{w}(b) = \{\hat{w}_i(b)\}_i$, compute the excess labor demand, $Z_i(\hat{w}(b))$, for every $i \neq \text{US}$ using (2). If $\max_{i \neq \text{US}} \{|Z_i(\hat{w}(b))|\} < \text{tol}$, then stop. Otherwise, compute a new guess for $i \neq \text{US}$:

$$\hat{w}_i(b+1) = \hat{w}_i(b) + \kappa^w Z_i(\hat{w}(b)) \quad (3)$$

where κ^w is a positive constant. Intuitively, this parameter controls by how much the algorithm increases the wage of countries with demand above supply (i.e., $Z_i(\hat{w}(b)) > 0$). The algorithm converges for κ^w small enough. It should converge in just a few minutes in any computer. You can evaluate if it is converging by checking whether the maximum error in excess demand is shrinking in each step b .

- iv. For the shock in item 1.iii, report the changes in the real wage ($\hat{w}_i / \hat{P}_i$) and real expenditure ($\hat{X}_i / \hat{P}_i$) of each country. For the sample of countries excluding China, run separate regressions of the log change in the real wage ($\ln \hat{w}_i / \hat{P}_i$) and the real expenditure ($\ln \hat{X}_i / \hat{P}_i$) on each of the following variables: the change in domestic trade share ($\ln \hat{\pi}_{ii}$), the change in market access ($\ln \hat{\Phi}_i$), and the initial spending share on Chinese goods ($\pi_{i\text{China}}^0$). Provide a brief intuition for your findings.
- v. We now consider the impact of the same shock introduced in item 1.iii, but under the assumption of a different numeraire of international transfers, while maintaining the US wage as the world economy's numeraire. Specifically, assume that transfers are set in terms of the Chinese wage $h(w) = w_{\text{China}}$, so that $\frac{h(w^0 \hat{w})}{h(w^0)} = \hat{w}_{\text{China}}$ and set the US wage to one, $w_{\text{US}}^0 = \hat{w}_{\text{US}} = 1$. Compare the implied changes in the real wage and real expenditure to those obtained in item 1.iii. Provide a brief explanation for why some countries gain more, while others gain less. Your answer should be backed by features of the data.